SC Algebra 1

# Graphing in Slope-Intercept Form

Name: _____ Period: ___

*Notes • y = mx + b*

## y = mx + b

**m** is the _____: _____. How steep, and which way.

**b** is the _____: the point (0, b), where the line crosses the _____. Where you START.

A whole-number slope is a fraction over 1: m = −2 means −2/1: down 2, right 1.

## 1. Find m and b

| y = 3x − 5 | y = −x + 4 | y = (2/3)x + 1 | y = 5 − 2x | y = 4x | y = −3 |
|---|---|---|---|---|---|
| m = _____<br>b = _____ | m = _____<br>b = _____ | m = _____<br>b = _____ | m = _____<br>b = _____ | m = _____<br>b = _____ | m = _____<br>b = _____ |

**Watch out:** y = 5 − 2x is y = −2x + 5. The slope is the number WITH the x (and its sign). y = −3 has no x: the slope is 0.

## 2. Graph it in 3 steps

| 1. START | 2. MOVE | 3. CONNECT |
|---|---|---|
| Plot b on the y-axis: (0, b). | Rise (up +, down −), then run RIGHT. Plot. Do it again. | Ruler line through the points, arrows on both ends. Check: m + goes uphill, m − goes downhill. |

## 3. Examples

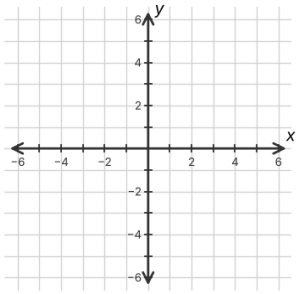
**Example 1**

**y = 2x − 3**

Start: b = ____ → plot (0, ____)

Move: m = ____ → rise ____, run ____

Points: _____



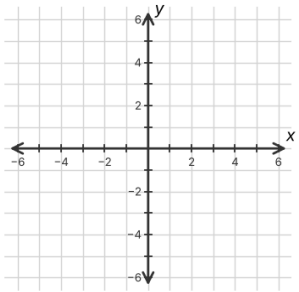
**Example 2 (fraction slope)**

**y = −(3/4)x + 2**

Start: b = ____ → plot (0, ____)

Move: m = ____ → rise ____, run ____

Points: _____



**Example 3 (rewrite first!)**

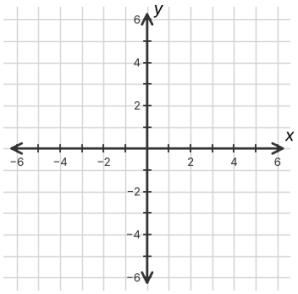
**y = −x + 5**

Rewrite: y = _____

Start: b = ____ → plot (0, ____)

Move: m = ____ → rise ____, run ____

Points: _____



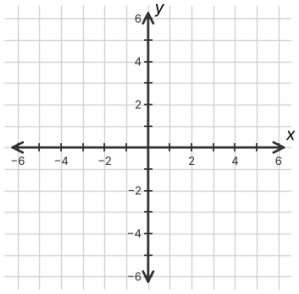
**Example 4 (b = 0)**

**y = (1/2)x**

Start: b = ____ → plot (0, ____)

Move: m = ____ → rise ____, run ____

Points: _____

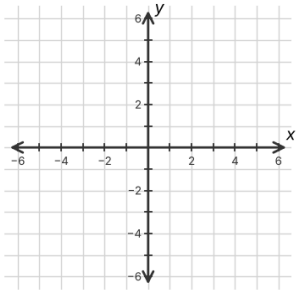


**Special lines**

y = 3: slope = ___ → _____

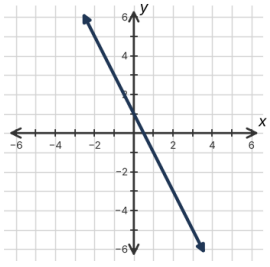
x = −2: slope is _____ → _____

*x = −2 has no y, so it can't be written as y = mx + b.*



**4. Write the equation from a graph**

Read b where the line crosses the y-axis. Count rise/run from that point to the next lattice point.



b = ____ m = ____ → _____

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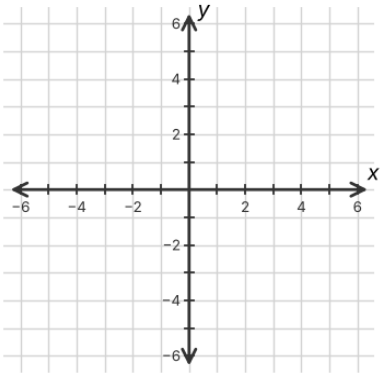


## 5. You try

**Graph**

**$y = -(1/3)x + 2$**

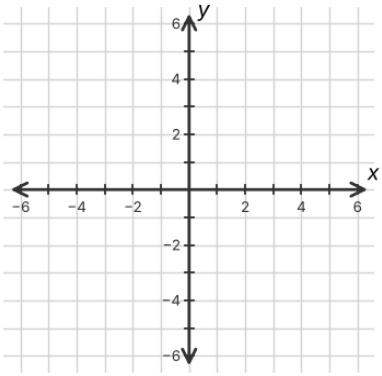
b = ____ m = ____



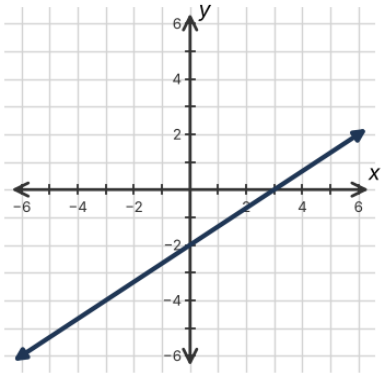
**Graph**

**$y = 3x - 4$**

b = ____ m = ____



**Write the equation**



b = ____ m = ____

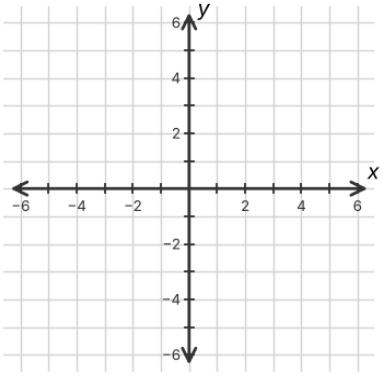
y = _____

**Graph**

**$y = 4 - x$**

Rewrite: y = _____

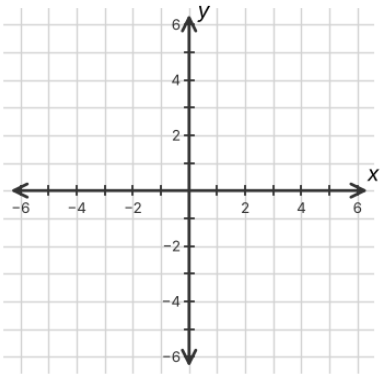
b = ____ m = ____



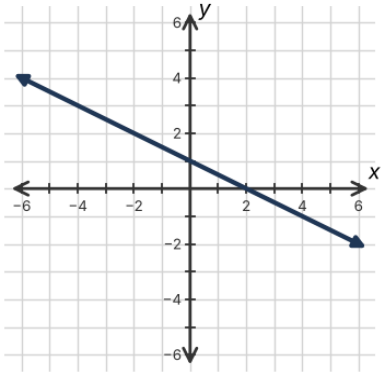
**Graph**

**$y = (2/5)x - 3$**

b = ____ m = ____



**Write the equation**



b = ____ m = ____

y = _____

**Watch out:**

× Starting on the x-axis. b is on the y-axis. × Running LEFT for a negative slope: put the negative on the rise (down), keep the run right. × Reading y = 5 − 2x as slope 5.

**★ Think deeper:** Graph $y = 2x + 1$ and $y = 2x - 3$ on the same grid. What do you notice? Why does that happen?

_____

